

REVIEW ARTICLE

Artificial intelligence in hepatocellular carcinoma diagnosis: a comprehensive review of current literature

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Key words

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Abstract

Background and Aim: Hepatocellular carcinoma (HCC) diagnosis mainly relies on its pathognomonic radiological profile, obviating the need for biopsy. The project of incorporating artificial intelligence (AI) techniques in HCC aims to improve the performance of image recognition. Herein, we thoroughly analyze and evaluate proposed AI models in the field of HCC diagnosis.

Methods: A comprehensive review of the literature was performed utilizing MEDLINE/PubMed and Web of Science databases with the end of search date being the 30th of September 2023. The MESH terms “Artificial Intelligence,” “Liver Cancer,” “Hepatocellular Carcinoma,” “Machine Learning,” and “Deep Learning” were searched in the title and/or abstract. All references of the obtained articles were also evaluated for any additional information.

Results: Our search resulted in 183 studies meeting our inclusion criteria. Across all diagnostic modalities, reported area under the curve (AUC) of most developed models surpassed 0.900. A B-mode US and a contrast-enhanced US model achieved AUCs of 0.947 and 0.957, respectively. Regarding the more challenging task of HCC diagnosis, a 2021 deep learning model, trained with CT scans, classified hepatic malignant lesions with an AUC of 0.986. Finally, a MRI machine learning model developed in 2021 displayed an AUC of 0.975 when differentiating small HCCs from benign lesions, while another MRI-based model achieved HCC diagnosis with an AUC of 0.970.

Conclusions: AI tools may lead to significant improvement in diagnostic management of HCC. Many models fared better or comparable to experienced radiologists while proving capable of elevating radiologists’ accuracy, demonstrating promising results for AI implementation in HCC-related diagnostic tasks.

Introduction

The concept of artificial intelligence (AI) was initially introduced in the 1950s and has experienced tremendous development in recent years with numerous applications in the field of medicine. It is currently used for diagnostic, therapeutic, and prognostic purposes, allowing for the effective analysis and use of the vast amount of digital data in the clinical setting.¹ AI can leverage available data through machine learning (ML) and deep learning (DL) techniques to contribute to decision-making and reduce the potential for human error. ML is automated learning from a set of previously provided data utilized to train a model in organizing and recognizing specific patterns. As for DL, it is a more extensive instance of learning, attempting to simulate the workings of the human brain, thus requiring larger data and more complex algorithms.²

Hepatocellular carcinoma (HCC) is the most common primary liver malignancy, ranks fifth among the most prevalent

malignancies and second as a cause of death due to cancer worldwide.³ HCC accounts for approximately 90% of primary liver cancers, showing a universally increasing incidence rate. In the vast majority of cases, HCC develops in patients with underlying cirrhosis, and about one third of cirrhotic patients will eventually develop HCC.³ A typical HCC lesion, according to the European Association for the Study of the Liver (EASL), will be larger than 1 cm on a cirrhotic liver and will show arterial phase hyperenhancement with washout at portal or delayed phases.^{4,5}

The utilization of AI in cases of HCC has attracted substantial interest, due to its high incidence and characteristic radiologic profile, which allows for diagnosis and treatment initiation without the need for invasive diagnostic techniques.⁶ HCC is unique since it possesses a quasi-pathognomonic imaging phenotype that can allow for treatment initiation without any further invasive diagnostic tests.¹ The integration of hemodynamic parameters from

multiple phase images is required for HCC diagnosis, making AI assistance necessary. AI has demonstrated its ability to support various stages of HCC diagnosis by analyzing clinical, radiological, histological, and biomarker data.¹ Radiomics focuses on the application of ML techniques in the field of radiology. The main goal is to extract high-dimensional features from imaging data, providing quantitative information about the tumor behavior and heterogeneity profiling. The traditional radiomics workflow entails a number of steps, from image acquisition and feature extraction to feature selection and model construction.⁷

The objective of our review is to critically assess and analyze published AI models focusing on various diagnostic modalities used in HCC diagnosis by appraising their methodology, evaluating the quality and quantity of data used, checking their generalizability, and, rather importantly, exploring whether they can substantially enhance the efficiency of the currently used diagnostic algorithm.

Methodology and materials

A comprehensive review of the literature was performed utilizing MEDLINE/PubMed and Web of Science databases with the end of search date being the 30th of September 2023. The MESH terms “Artificial Intelligence,” “Liver Cancer,” “Hepatocellular Carcinoma,” “Machine Learning,” and “Deep Learning” were searched in the title and/or abstract. The references of relevant articles were reviewed to identify additional eligible publications. An expert review of the eligible literature was performed, and the most relevant and informative citations were identified for inclusion. Literature search was completed in September 2023. Study inclusion was performed by two researchers (OPC and CL) separately, and results were cross-validated. Included studies were published between 2017 and 2023 and explored a variety of diagnostic modalities, including ultrasound (US), computed tomography (CT), and magnetic resonance imaging (MRI) (Figs 1, 2). Then one researcher proceeded with data extraction, focusing on methods and results of each study.

Search criteria. The following questions defined the shape of our review:

- 1 Which are the most successful AI models across all modalities for HCC diagnosis?

- 2 What are the characteristics of models achieving high accuracy and AUC in HCC diagnosis?
- 3 How can AI models assist clinicians in detecting HCC, and how can they seamlessly be incorporated in the clinical setting?
- 4 What are the current limitations utilizing AI models in the clinical setting?

AI glossary. AI is a branch of computer science that entails the research and construction of mathematical models, granting machines the ability to reason and perform certain functions, including problem-solving, object and word recognition, and even decision-making.² ML is a subfield of AI that utilizes data to produce algorithms that can detect certain behavioral patterns and build predictive models. ML aims to create a model that exploits statistical dependencies and correlations in a dataset, without being explicitly programmed. This process is divided into two phases. During the training phase, the model is exposed to a subset of available data (training dataset). During the second phase, the model is tested on the other subset of the dataset (test dataset) in order to evaluate its ability to generalize its training performance over unseen data. Examples of ML techniques include, among others, support vector machines (SVM), artificial neural networks (ANNs), or classification and regression trees. ANNs are algorithms, inspired by biological neurons, that consist of several neuron-containing layers. The first layer is the input layer, while the last one is the output layer.² More recently, the development of DL, a subfield of ML, has allowed for the creation of multilayered deep neural networks (DNNs) with more than one hidden layers. DL models understand and learn the most characteristic radiologic features directly from raw images pixels, hence overcoming the subjective input that is required during the engineering of machine learning models.⁸ From all DNNs, the convolutional neural network (CNN) has proven to possess great value in specific pattern recognition and, hence, the evaluation of radiologic images.^{1,9} Unlike other ML methods, CNNs do not require predefined specific features to learn image interpretation, since they can automatically extract these features through backpropagation using multiple layers.¹⁰ A representation of the utilization of ML and DL techniques in the included studies has been provided in Figure 3.

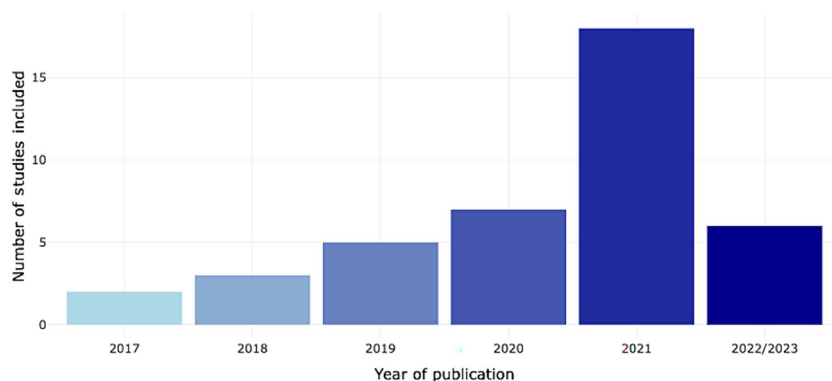


Figure 1 Number of included studies per year of publication.

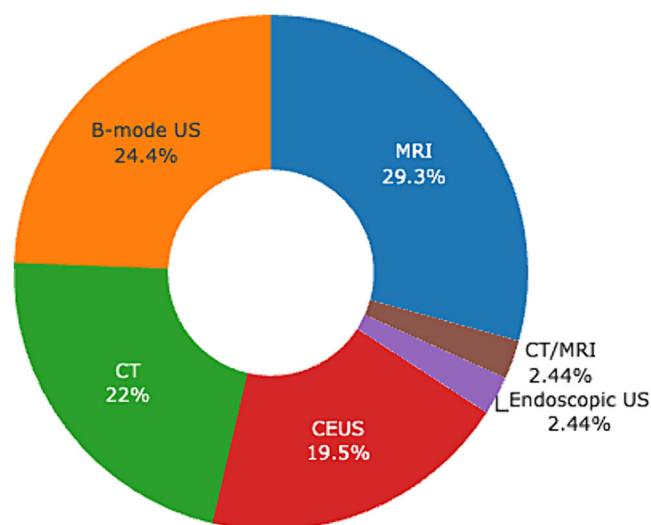


Figure 2 Percentage of diagnostic modalities utilized in AI diagnostic models for HCC. ■, MRI; ■, B-mode US; ■, CT; ■, CEUS; ■, Endoscopic US; ■, CT/MRI.

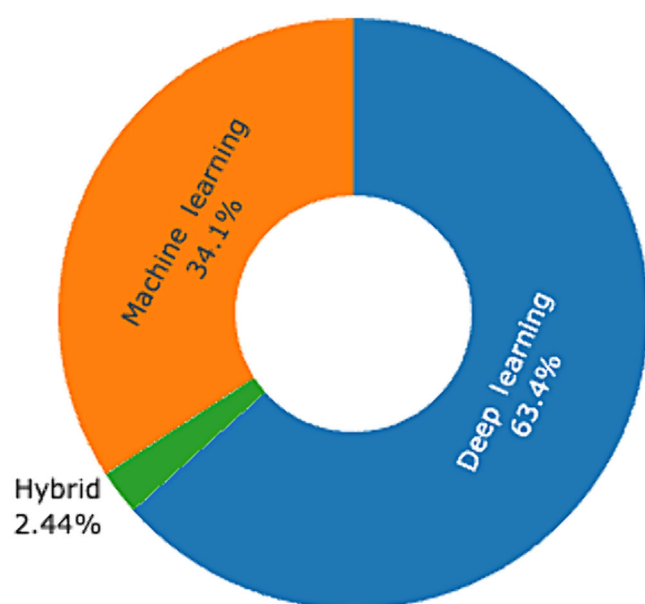


Figure 3 Percentage of AI models using machine learning, deep learning or hybrid techniques. ■, Deep learning; ■, Machine learning; ■, Hybrid.

Results

AI and B-mode US. AI models based on B-mode US images can aid even inexperienced radiologists during diagnostic work-up of focal hepatic masses. Misdiagnoses are common in B-mode US interpretation, as it relies on real-time judgment, based on individual visual perception.⁹ A total of three studies^{11–13} focused on the differentiation of focal liver lesions as benign or possibly malignant (Table 1). Yang *et al.* developed a deep CNN using 24,343 B-mode US and reported an AUC of 0.924, 86.5% sensitive and 85.5% specific, surpassing the performance of radiologists with over 15 years of experience. The main drawback of this study was the exclusion of cystic, small, and histologically indeterminate lesions, as a wider variety of lesion types could decrease accuracy. The more challenging task of classifying liver lesions among possible types was attempted by four studies.^{15,18,19,27} In a 2021 large

multicenter study,¹⁸ a CNN model was trained with US data from 3,487 patients and achieved 87% (95% CI: 84.3–89.6) and 75% (95% CI: 71.7–78.3%) detection rate in an internal validation (IV) and EV cohorts, respectively, on the task of classification among HCCs, cysts, hemangiomas, focal fatty sparing, and focal fatty infiltration. HCC diagnosis was achieved with specificity and negative predictive value (NPV) of 94.4% (95% CI: 92.8–96.0) and 97.4% (95% CI: 96.2–98.5), respectively, in the EV set. Nishida *et al.*¹⁵ constructed a CNN model with 70,950 still B-mode images and classified liver lesions among cysts, hemangiomas, HCCs, and metastatic tumors. Accuracy for malignancy was 94.3% and for classification 91.1%. The model outperformed human experts in an independent external test cohort of video images (89.1% vs 67.3%). This CNN displayed higher overall accuracies than contrast-enhanced US (CEUS) models, which may be accredited to the vast amount of data used for its training. As for

Table 1 AI studies based on ultrasound pertaining to HCC diagnosis

Reference	Year	Aim of the study of the use of AI in imaging techniques	Diagnostic tool	AI tool used	Database size	AUC (best value)	Acc/Sn/Sp (best values)
Zhang <i>et al.</i> ¹⁴	2022	Diagnosis of HCC in HBV-infected liver	B-mode US	DL	407 patients	0.936	86.4%/96%/76.9%
Nishida <i>et al.</i> ¹⁵	2022	FLL classification	B-mode US	DL/CNN	70,950 images (model 3)	—	93.4%/67.5%/96% (HCC diagnosis)
Liu <i>et al.</i> ¹⁶	2022	Benign vs malignant	CEUS	DL	303 patients	0.957	94%/96.6%/90.5%
Turco <i>et al.</i> ¹⁷	2022	Benign vs malignant	CEUS	ML	72 patients; 87 images	0.840	84%/76%/92%
Tiyarattanachai <i>et al.</i> ¹⁸	2021	FLL classification	B-mode US	DL/CNN	3,847 patients; 40,397 images	—	94.9%/73.6%/97.8%
Ryu <i>et al.</i> ¹⁹	2021	FLL classification/joint segmentation	B-mode US	DL	3,873 patients; 4,309 images	0.947	82.2%/86.7%/98.7%
Ren <i>et al.</i> ²⁰	2021	Differentiation between HCC and ICC	B-mode US	ML/SVM	226 patients	0.874	87.2%/88.9%/86.7%
Mayya <i>et al.</i> ²¹	2021	Benign vs malignant	Endoscopic US	DL/CNN	210,685 images; 256 patients	0.861 (image) 0.904 (video)	—/90%/71% (image) —/100%/80% (video)
Xi <i>et al.</i> ¹²	2021	Benign vs malignant	B-mode US	DL/CNN	596 patients; 911 images	0.860	84%/87%/78%
Li <i>et al.</i> ²²	2021	aHCC vs FNH	CEUS	ML/SVM	226 patients	0.930	—/93.5%/84.9%
Căleanu <i>et al.</i> ²³	2021	FLL classification	CEUS	DL	91 patients; 273 video files	—	88%/—/—
Hu <i>et al.</i> ⁸	2021	Benign vs malignant	CEUS	DL/CNN	363 patients	0.934	91%/92.7%/85.1%
Huang <i>et al.</i> ²⁴	2020	aHCC vs FNH	CEUS	ML/SVM	257 images	0.944	—/94.76%/93.62%
Yang <i>et al.</i> ¹¹	2020	Benign vs malignant	B-mode US	DL/CNN	2143 patients; 24,343 images	0.924	84.7%/85.5%/85.5%
Schmauch <i>et al.</i> ¹³	2019	Benign vs malignant	B-mode US	DL	367 images	0.891	—/—/—
Guo <i>et al.</i> ²⁵	2018	Benign vs malignant	CEUS	ML	93 patients	0.953	90.4%/93.6%/86.9%
Bharti <i>et al.</i> ²⁶	2018	Classification (normal liver, chronic liver disease, cirrhosis, HCC over cirrhosis)	B-mode US	CNN/SVM	94 patients; 189 images	—	96.6%/96.9%/99.8%
Hassan <i>et al.</i> ²⁷	2017	FLL classification	B-mode US	DL/ANN	110 images	—	97.2%/98%/95.7%
Kondo <i>et al.</i> ²⁸	2017	Benign vs malignant/FLL classification	CEUS	ML/SVM	98 patients	—	91.8%/94%/87.1% (benign vs malignant)

ANN, artificial neural network; CEUS, contrast-enhanced ultrasound; CNN, convolutional neural network; DL, deep learning; FLL, focal liver lesion; FNH, focal nodular hyperplasia; HCC, hepatocellular carcinoma; ICC, intrahepatic cholangiocarcinoma; ML, machine learning; SVM, support vector machine; US, ultrasound.

Ryu *et al.*,¹⁹ the proposed CNN had the joint purpose of segmentation and classification of FLLs among four lesion types. Trained and tested with 4,309 US images, this model displayed a mean accuracy of 82.2% for FLL classification, while the evaluation of segmentation was based on the Jaccard index (JI), with a reported value of 68.5%. Finally, three other B-mode US models were identified, focusing on: diagnosis of HCC in HBV-infected livers¹⁴; recognizing normal liver parenchyma, chronic liver disease, cirrhosis and HCC evolved on cirrhotic livers²⁶; and differentiating between HCC and intrahepatic cholangiocarcinoma.²⁰

AI and CEUS. CEUS generally outperforms B-mode US in diagnosing FLLs and HCC, specifically, and AI has augmented its capabilities in identifying potential malignancies. The task of differentiating benign and malignant FLLs was tackled by five studies^{8,16,17,21,25} (Table 1). Hu *et al.*⁸ constructed a CNN model trained on four-phase CEUS video analysis data from 363 patients. They achieved an accuracy of 91% and an AUC of 0.934 on the testing set, slightly outperforming resident radiologists and matching experts. Due to the AI model's higher positive predictive value (PPV) when compared to residents (95.6% vs 88–6–89.7%), an AI-assistance strategy was developed, improving the sensitivity and accuracy of resident radiologists on a par with the level of the experts. Clinical parameters were incorporated by Liu *et al.*,¹⁶ who proposed a CEUS DL model to detect malignancy. Three hundred three patients were divided into training, IV, and EV cohorts. The combined CEUS-clinical model was the most successful, reporting AUCs of 0.969 and 0.957 and accuracies of 96% and 94% in the IV and EV groups, respectively, outperforming two radiologists, while processing information two times faster. For the classification of FLLs,^{23,28} Căleanu *et al.* used CEUS data to develop a DL model aiming to classify FLLs among five types, achieving a general accuracy of 88%, in a 91 patient cohort with 30 HCC cases. Li *et al.*²² investigated the capability of ML to elevate the diagnostic accuracy when differentiating between atypical HCCs (aHCCs) and focal nodular hyperplasia (FNH). Two hundred twenty-six FLL three-phase CEUS images were processed and generated an ultrasomics signature. This model, a radiologist's score, and their combination were used to train, via SVM, three distinct versions of a predictive model. The combined model achieved the highest AUC of 0.930 in aHCC versus FNH differentiation (vs 0.860 and 0.840 for the ultrasomics model and the radiologist, respectively). This goal was shared with Huang *et al.*,²⁴ who reported an even higher AUC of 0.944, in addition to 94.76% sensitivity and 93.62% specificity.

AI and endoscopic US. An exciting prospect in the field of AI was introduced by Marya *et al.*,²¹ who were the first to develop an endoscopic US (EUS)-based CNN for the identification and classification of FLLs. Still images and videos of liver tissue and FLLs from 256 patients trained the model, which was deemed 92% successful in autonomously detecting FLLs using occlusion heatmap analysis of video assets. As for classification of malignant FLLs, the reported AUC of the CNN model was 0.861 in still images and 0.904 in full-length video assets, also achieving 100% sensitivity in the video test set.

AI and CT. US-detected liver lesions usually require further evaluation for classification with dynamic contrast-enhanced axial imaging. The Liver Imaging Reporting and Data System (LI-RADS) criteria apply to CT or magnetic resonance imaging images, stratifying FLLs in five categories, ranging from definitely benign to definitely HCC. Three studies focused on the task of HCC diagnosis^{4,29,30} (Table 2). Using CT data from 7,512 patients, a model was engineered by Wang *et al.*, evaluated in IV (385 patients) and EV (556 patients) cohorts. HCC diagnosis was achieved with an AUC of 0.887 and 0.883 on the IV and EV set, respectively. High accuracy was reported (IV: 81%, EV: 81.3%). The HCC risk score was evaluated after patients from both cohorts underwent surgical resection. The model predicted remarkably higher HCC risk scores for patients with multiple tumors and severe fibrosis or cirrhosis and performed better or equally to three radiologists on 177 random cases from both validation sets. To assess the model's performance as an aid, the latter performed HCC diagnosis on the same dataset 12 months later, assisted by the model. The mean accuracy increased significantly, from 0.818 to 0.873 in the IV set and from 0.793 to 0.854 in the EV set, and when tested on 92 new cases, mean accuracies were 0.826 and 0.820 on the IV and EV cohorts, respectively. Kim *et al.* constructed a CT-based algorithm for detecting primary malignant FLLs in patients at increased risk for HCC development. The database included 1,350 multiphase CT scans, ~56% of which were used for training and tuning purposes and the rest were assigned to the test set. Specifically for HCC, the reported sensitivity was 85.6%, broken down to 95.9% in >20 mm, 74.7% in 10–20 mm, and 33.3% in <10 mm malignant lesions.

The purpose of FLL classification was explored by three studies^{31,35,36} (Table 2). Gao *et al.* explored the combination of multi-phase contrast-enhanced CT and clinical parameters. Seven hundred twenty-three patients with malignant liver tumors from two centers (HCC, ICC, and metastatic lesions) were included. CT scans from one center consisted the training and IV cohorts, and from the other center, the EV cohort. The model had 86.2% accuracy and an AUC of 0.893 for differentiating between HCC and ICC in the IV set and a 72.6% accuracy among HCC, ICC, and metastasis, comparable to the doctors' consensus (70.8%). Specifically for HCC diagnosis, AUC was 0.937. As an aid, the model managed to increase the doctors' accuracy by 8.3%. The generalizability was supported by an accuracy of 82.9% in the EV set, in addition to 98% sensitivity and 91.8% specificity for HCC diagnosis. Yang *et al.*³² designed a CT-based CNN aiming to simultaneously detect and segment abnormal liver densities. For training, 394 patients' radiologic data were acquired from one center and 337 patients from a second center consisted the testing set. The model achieved an AUC of 0.949 for detection of abnormal FLLs and a mean Dice coefficient of 0.797 for its segmentation ability. DL models were also proposed for differentiating between HCC and ICC³³ and comparing the diagnostic power of three-phase versus four-phase CT in HCC patients.³⁴

AI and MRI. A total of three studies focused on HCC diagnosis^{37–39} (Table 3). A fully automated CNN was introduced by Kim *et al.*, trained on MRI data from 549 patients. In the IV test set, the model reported an AUC of 0.970, being 94% sensitive and 99% specific. In a 45-patient EV cohort, it achieved a sensitivity

Table 2 AI studies based on computed tomography pertaining to HCC diagnosis

Reference	Year	Aim of the study of the use of AI in imaging techniques	Diagnostic tool	AI tool used	Database size	AUC	Acc/Sn/Sp
Gao <i>et al.</i> ³¹	2021	Classification of malignancy	CT	DL/CNN/RNN	723 patients	0.986 (HCC diagnosis)	83%/—/—
Yang <i>et al.</i> ³²	2021	Detection and segmentation (abnormal liver density)	CT	DL/CNN	394 patients; 10,130 images	0.949	—/—/—
Kim <i>et al.</i> ³⁰	2021	HCC diagnosis	CT	DL/CNN	1,320 patients; 1,350 images	—	—/86%/—
Nakai <i>et al.</i> ³³	2021	HCC vs ICC	CT	DL/CNN	617 patients	0.870	61%/75%/88%
Wang <i>et al.</i> ²⁹	2021	HCC diagnosis	CT	DL/deep residual CNN	7,512 patients	0.883	81%/89%/74%
Shi <i>et al.</i> ³⁴	2020	HCC diagnosis (three-phase vs four-phase CT)	CT	DL/CDN	342 patients; 449 FLLs	0.925 (four-phase) 0.920 (three-phase/no pre-contrast)	83%/92%/77% (four-phase) 86%/74%/94% (three-phase/no pre-contrast)
Cao <i>et al.</i> ³⁵	2020	FLL classification	CT	DL/CDN	517 FLLs	0.920 (HCC diagnosis)	91.6%/—/96.4% (HCC diagnosis)
Mokrane <i>et al.</i> ⁴	2020	HCC diagnosis	CT	ML	178 patients	0.660	—/70%/54%
Yasaka <i>et al.</i> ³⁶	2018	Benign vs malignant/FLL classification	CT	DL/CNN	460 patients; 1,068 images	0.920 (benign vs malignant)	84%/71%/— (HCC diagnosis)

CDN, convolutional dense network; CNN, convolutional neural network; CT, computerized tomography; DL, deep learning; FLL, focal liver lesion; HCC, hepatocellular carcinoma; ICC, intrahepatic cholangiocarcinoma; ML, machine learning; RNN, residual neural network.

Table 3 AI studies based on magnetic resonance imaging pertaining to HCC diagnosis

Reference	Year	Aim of the study of the use of AI in imaging techniques	Diagnostic tool	AI tool used	Database size	AUC	Acc/Sn/Sp
Sun <i>et al.</i> ³⁹	2022	HCC diagnosis	MRI	ML/SVM	124 patients	0.970	94.1%/94.1%/94.1%
Hu <i>et al.</i> ⁷	2022	HCC vs ICC	MRI	ML/Tree-Based Optimization Tool	489 patients	0.760–0.790	73–75%/65–75%/75–79%
Liu <i>et al.</i> ⁴⁰	2021	HCC vs non-HCC	CT/MRI	ML/SVM	85 patients	0.790–0.810 (MRI) 0.810 (CT)	—/68%/88% (MRI)
Stollmayer <i>et al.</i> ⁴¹	2021	FLL classification	MRI	DL/densely connected CNN	216 images	0.978 (2D model) 0.943 (3D model)	—/93%/97% (2D-model) —/83%/92% (3D-model)
Zhong <i>et al.</i> ⁴²	2021	Small HCC vs benign lesions	MRI	ML/logistic regression	150 patients	0.975	97.4%/97.3%/97.7%
Oestmann <i>et al.</i> ¹⁰	2021	HCC vs non-HCC	MRI	DL/CNN	118 patients; 150 lesions	0.912	87.3%/92.7%/82%
Bousabarah <i>et al.</i> ³⁸	2021	HCC diagnosis	MRI	DL/CNN	174 patients; 231 lesions	—	—/75%/—
Kim <i>et al.</i> ³⁷	2020	HCC diagnosis	MRI	DL/CNN	549 patients	0.900	94%/87%/93%
Zhen <i>et al.</i> ⁴³	2020	FLL classification	MRI	DL/CNN	1,210 patients	0.925 (HCC diagnosis)	—/87.2%/91.6% (HCC diagnosis)
Jiang <i>et al.</i> ⁴⁴	2019	HCC diagnosis (radiomics vs LI-RADS vs 2018 EASL)	MRI	ML/logistic regression analysis	211 patients; 229 lesions	0.810	—/73%/71%
Wu <i>et al.</i> ⁴⁵	2019	HCC vs hepatic hemangioma	MRI	ML	369 patients; 446 lesions	0.890	—/82.2%/71.4%
Hamm <i>et al.</i> ⁴⁶	2019	FLL classification	MRI	DL/CNN	494 lesions	0.992 (HCC diagnosis)	92%/92%/98% (overall)
Jansen <i>et al.</i> ⁴⁷	2019	FLL classification	MRI	ML	95 patients; 213 lesions	0.910 (HCC diagnosis)	—/94%/98% (HCC diagnosis) 88%/73%/56% (HCC diagnosis)

CNN, convolutional neural network; CT, computerized tomography; DL, deep learning; EASL, European Association for the Study of the Liver; FLL, focal liver lesion; HCC, hepatocellular carcinoma; ICC, intrahepatic cholangiocarcinoma; LI-RADS: Liver Imaging Reporting and Data System; ML, machine learning; MRI, magnetic resonance imaging; SVM, support vector machine.

and specificity of 87% and 93%, respectively, and an AUC of 0.900. Lesions missed by less experienced radiologists, but detected by the CNN, had a mean size of 0.8–1.2 cm, signifying an advantage in smaller lesion detection. Notably, entire phase MR images were the input, while other DL models require complicated and time-consuming manual image preprocessing. However, a false positive issue was introduced, as intrahepatic vessels, the gallbladder, and the heart were misdiagnosed for HCC. As these mistakes can easily be detected by radiologists, the model's capability as an aid is not substantially downgraded. The task of FLL classification was taken on by four studies^{41,43,46,47} (Table 3). Zhen *et al.* developed a CNN incorporating both unenhanced and enhanced MRI data, in addition to clinical parameters. One thousand and two hundred ten patients with FLLs were included, and three separate signatures were developed, including a seven-way classifier, a binary classifier, and a three-way malignant classifier. The EV cohort consisted of 201 patients. In the seven-way classification, the CNN model reported AUCs ranging from 0.897 to 0.987 and performed comparably to the radiologists. For binary classification, malignancy diagnosis had an AUC of 0.946 using a signature with three unenhanced MRI sequences. The combined signature of unenhanced imaging and clinical data detected HCC among other malignancies with an AUC of 0.985, outperforming three radiologists, who were aware of relevant clinical data and had assessed all available sequences.

For the goal of classification among six types and among three broader LI-RADS categories, Hamm *et al.* trained an MRI-based CNN with 494 FLLs constituting the training and validation sets. The model demonstrated an accuracy of 91.9% for individual lesions and 94.3% for the broader categories, achieving 90% sensitivity and 98% specificity, respectively, while the average values of specialists were 82.5% and 96.5%, respectively. Specifically for HCC diagnosis, sensitivity was 90% and AUC was 0.992. The model performed well in cases misclassified by radiologists. Four of the included studies explored the use of AI in discerning HCC from other malignancies,^{7,10,40,45} or benign tumors. Hu *et al.* used an automated Tree-Based Optimization Tool (TPOT) to differentiate HCC from ICC. Five hundred fourteen patients with lesions bigger than 2 cm in all dimensions and well-demarcated borders were included. There were 24 lesions defined as LR-M (probable or definite malignancy) based on LI-RADS in the test set. Four three- to eight-year experienced radiologists classified the cases as HCC or ICC, blinded to histopathological data. The model had an AUC of 0.760–0.790 and 73–75% accuracy, 65–75% sensitivity, and 75–79% specificity on the test set but had an AUC of 0.600 and accuracy of 63% on the LR-M subset, below that of radiologists (81%). Other approaches utilizing AI with MRI included an ML model⁴² aiming to differentiate between small HCC and benign tumors on cirrhotic livers and a comparison of the performance of an ML model against LI-RADS and 2018 EASL criteria in HCC diagnosis in high-risk patients.⁴⁴

AI and histopathology. A valuable AI assistant for histopathological classification must be easily accessible, quick, and accepting of various types of input data. Kiani *et al.*⁴⁸ presented such a DL model to assist pathologists in differentiating HCC and ICC. Seventy hematoxylin and eosin-stained WSI were used

for training, and the model's performance was compared with 11 radiologists with varying experience levels. Accuracy was 88.5% in the IV and 84.2% on the EV set. Although it did not manage to alter the mean accuracy of all pathologists in a statistically significant way, it elevated the diagnostic performance in a subgroup that excluded two pathologists without well-defined experience level. All pathologists' accuracy was individually affected by the AI model's predictions, either correct or incorrect, raising concern over unintended negative consequences when implementing AI models in the clinical setting. A residual CNN, developed by Feng *et al.*,⁴⁹ aimed to minimize manual processing in HCC classification. Hematoxylin and eosin-stained histopathological slides were obtained from 592 HCC patients. The model achieved pixel-level accuracy and slide-level accuracy of 87.8% and 98.7%, respectively, on the IV test, in addition to 87.9% accuracy on an EV set of 157 HCC slides.

AI and biomarkers. Cheng *et al.*⁵⁰ based their model on personalized biological pathways analysis and the ML method of regularized regression, achieving high accuracy for HCC diagnosis. Ten thousand and four hundred one shared genes were assigned to two training sets and an EV set. The model reported an AUC of 0.992 and 84.9% sensitivity, displaying high accuracy and transferability. Eight hundred twenty-four characteristic genes were identified, 33 of which correlated either positively or negatively with overall survival. Via regression analysis, 12 characteristic genes created a risk signature that achieved high efficiency and stability of prognosis prediction. A novel ML signature was developed by Sato *et al.*⁵¹ for HCC prediction, focusing on the model's adaptability to the clinical setting and the use of real-world data. One thousand and five hundred eight-two patients, with the ratio of HCC to non-HCC being roughly 1:2. Gradient boosting was the most successful ML classifier, achieving an HCC predictive accuracy of 87.3%, an AUC of 0.940, 93.3% sensitivity, and 76% specificity. It also accomplished to decrease misclassification rate by roughly 50% when compared to the performance of single tumor markers.

Discussion

The consensus diagnostic approach in HCC considers tumor-specific characteristics acquired via imaging modalities, such as B-mode US, CEUS, CT, and MRI, as well as histopathological slides and tumor marker levels. Current hurdles include the subjectivity of interpreting these features among radiologists of different experience levels and the fact that there is a wide range of radiologic features characterizing HCCs, many of which are shared by other FLLs, benign or malignant. In an attempt to overcome these obstacles, AI models were introduced, to assist in the areas of diagnosis and cancer management. There is indeed a wide variety of models based on ML and DL displaying great promise, although not yet extensively used in everyday clinical practice. Radiomics analysis manually or automatically extracts quantitative texture features from images that may correspond to histopathological characteristics of FLLs. Thus, this approach shows promising resolving power in various levels of the HCC diagnostic approach, including detection of malignancy, HCC diagnosis and

segmentation, differentiation among multiple FLL types, and prediction of histopathological grade.

The role of AI in healthcare has been increasingly explored in recent years. Various studies have examined potential AI applications with encouraging results, including cancerous cell detection in acute lymphoblastic leukemia, as well as COVID-19 variant identification and severity of infection estimation.^{52–54} Specifically in the realm of liver pathology, AI models have aimed to predict the incidence of nonalcoholic fatty liver disease and fibrosis and have managed to accurately predict HCC development in patients with chronic viral hepatitis.^{55,56} The model by Wong *et al.* managed an AUC of 0.842 and 0.844 in training and validation cohorts, respectively, outperforming various HCC risk scores. Finally, the task of liver tumor segmentation has been tackled by previously reported studies. Lyu *et al.* trained a weakly-supervised CNN using Couinaud segment annotations and achieved comparable results to fully-supervised models with comparably lower annotating efforts.⁵⁷

US has become a highly useful tool for diagnosing and planning treatment for various diseases due to its many advantages, namely, being a noninvasive, low-cost, portable modality, capable of providing real-time imaging. However, reliability concerns arise due to the fact that the accuracy of the results can vary depending on the operator. This issue could be addressed through the use of AI in order to standardize the interpretation of US findings and reduce subjectivity. This underlines the importance of continued development and widespread use of AI-assisted US in the clinical setting.⁹ With respect to focal liver lesions and HCC, US is considered one of the first-line diagnostic imaging modalities during initial work-up. AI provides an upgrade to normal human vision and understanding of US images, as it is superior in monolithic tumor morphology reflection, in addition to granular and radiological pattern recognition. Moreover, the detection of HCC on a background of cirrhosis is considered a challenging task even for experienced radiologists.¹⁸

Going a step forward, CEUS involves injecting a specialized contrast agent containing microbubbles that clarify the vascular supply in the area of interest by highlighting both large vessels and microcirculation. The microbubbles create harmonic echoes that can be detected by the transducer.⁵⁸ A standard regarding HCC diagnosis is the challenge that arises in cases of aHCCs, which do not comply with the standard CEUS enhancement pattern of hyperenhancement in the arterial phase, followed by wash-out in the portal venous and late phases. On the contrary, they display hyperenhancement/iso-enhancement in the portal venous and late phases, which is typical of most benign FLLs. This issue extends to CT too, as per Kim *et al.*,³⁰ where it was speculated that most cases of false negatives were due to atypical HCC contrast-enhancement patterns and small tumor size, while the false positives were either the result of faulty segmentation and registration or of the erroneous detection of nonmalignant FLLs.

CT is favored over MRI due to its wider availability and lower cost. However, liver nodules may still be inconclusive on CT and require biopsy for confirmation, as suggested by the EASL guidelines.⁵ On the other hand, the American Association for the Study of Liver Diseases (AASLD) recommends following-up with indeterminate tumors rather than performing a biopsy. This highlights the need for an automated diagnostic model that can assist physicians in hepatic tumor diagnosis, reducing observer

variability and increasing diagnostic accuracy.⁵⁹ Notably, a challenge arises when variable-sized HCCs present at different phases of CECT with blurred margins, homogeneous, heterogeneous, or poor enhancement, as central scars or necrosis, or even with an infiltrative pattern.

A purely radiological diagnosis of HCC remains a challenging task, leading to the introduction of standardized image analysis and reporting frameworks, such as the LI-RADS criteria. However, these have been steadily becoming more complex, underlining the need for computational assistance, in order to become seamlessly integrated in the clinical workflow. DL CNNs have been widely used for radiological classification, as they do not require manual extraction of features from imaging data and may even recognize differentiating features invisible to the human eye.

MRI generally presents higher sensitivity than CT, and in the context of FLL evaluation, gadoxetate disodium (gadolinium EOB DTPA, Gd-EOB-DTPA, and gadoteric acid) is a hepato-specific paramagnetic gadolinium-based contrast agent used for HCC diagnosis. Its superior sensitivity to other agents is due to the characteristic hypointense depiction of HCC in the hepatobiliary phase.⁴⁶ MRI possesses an elevated soft tissue contrast resolution that allows CNNs to detect and analyze a broader spectrum of radiologic features and offers a three-dimensional volumetric approach that combats the problem of dependence on manual slide selection. A common challenge arises when differentiating the common mass-forming subtype of ICC (mICC), which shares radiologic features with poorly differentiated, hypovascular HCC variants. Also, small mICCs may present as hypervascular lesions on the arterial phase and, thus, mimic HCC. LI-RADS criteria contain a category for non-HCC malignant FLLs, named LR-M, where typical ICCs are usually included. However, cases of atypical ICCs tend to be misdiagnosed as HCCs by the LI-RADS criteria.⁷

The rapid progression of scientific research and discovery has further prompted pathologists in continuous subspecialization, which logically impedes their ability to retain a high level of expertise regarding a wide range of histopathological specimen subtypes. During the assessment of a FLL, both the histopathological classification and the differentiation of the tumor strain affect treatment options and prognosis. This presents a challenge where AI could aid pathologists. However, many proposed AI models have depended on automated processing of whole-slide images (WSI), which hinders their broad use due to the requirement of a completely digital workflow. Even in selected centers that such models can be used, the digitization process is rather time consuming, especially for time-sensitive tasks like tumor classification.

To date, tumor markers have been mostly used either for surveillance or adjunctively in the diagnostic process of HCC, with serum alpha-fetoprotein (AFP) utilized the most. However, AFP has many drawbacks, including low sensitivity, even with a low cutoff point, and low specificity, since its increase is also associated with various other conditions and tumors. Generally, the use of a sole tumor marker for the diagnosis of a multifactorial entity, such as HCC, certainly presents a challenge. Novel biomarkers for HCC diagnosis have not been sufficiently tested and have not demonstrated satisfactory sensitivity. In this regard, ML can aid in the development of signatures that enhance the diagnostic performance of these biomarkers. An interesting development are AI models, whose diagnostic prediction takes into account gene pathway data

and could find a role in clinical pathological diagnosis or serve as a means of possible follow-up in cases that undergo gene-related therapy. New alterations in genetic pathways and prognostic features of HCC can be discovered, which could set the theme for future research in this area.⁵⁰ Such predictive models play a major role in personalized medicine, since they are based on patient-specific characteristic to estimate the probability of a clinical event, such as the development of HCC.

A number of current challenges that limit the widespread use of AI applications for HCC diagnosis need to be addressed. The need to regulate its use in the medical setting, as is the case for all medical devices, suggests a major challenge for AI. Both the Food and Drug Administration and European Commission have already stated their intent for future legislation. In addition, the proposed AI models and algorithms will continuously develop and change from the moment they become available in the market, raising issues regarding both safety and intellectual property.⁶⁰ The quality and quantity of data is vital in successful AI training; thus, a limitation arises due to the inherent biases and questionable quality of currently available databases, which commonly suffer from wrong labeling, lack of standardization, and missing data. Furthermore, many of the models have a significant issue with small sample sizes, and despite attempts at data augmentation, the datasets are often too small for effective CNS training. Satisfactory large datasets could be created via multi-center individual-participant data sharing, which on the one hand is an interesting prospect; however, on the other hand, it raises both ethical and patient-privacy concerns.⁶⁰ It must be noted that in studies where any amount of image reprocessing is performed manually by radiologists, reproducibility, and performance of the model certainly take a hit. To this end, combined automated tumor segmentation is vital. Finally, the issue of interpretability is of great interest and significance regarding the implementation of AI in the medical setting. Most currently available models offer limited insight into their decision-making process, a fact that significantly undermines the trust that medical professionals have towards such aids. In addition, as use of AI becomes more popular, it is certain that problems of liability will emerge, especially in cases where AI and doctors do not see eye to eye.⁶⁰ Finally, although effective models have been described in all diagnostic modalities analyzed in the current review, drawing comparisons is challenging, given the fundamental differences in databases, artificial networks, and conclusions of the various studies. All the aforementioned models require larger databases and more representative external validation sets in order to report more accurate performance values. The power of AI software analysis can use input data such as histopathological slides or biomarker levels to elevate their predictive accuracy in multiple tasks, including HCC diagnosis. The power of AI software analysis with that data could offer critical aid in diagnosis of HCC with high predictive accuracy.^{61,62}

In conclusion, the advancement of AI applications in the diagnosis of HCC and clinical practice necessitates larger prospective studies that compare the effectiveness of medical professionals assisted by AI versus those without such assistance. To obtain more accurate performance assessments and evaluate how AI can improve the diagnostic process, it is essential to include radiologists with different levels of experience. It is also critical to acclimate medical professionals to AI techniques and prepare them for its inevitable integration into the field of liver cancer diagnosis.

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